What is `Dataset.map()`?

`Dataset.map()` is a method that applies a function to each example in a dataset.

Example:

dataset.map(function)

Here, the function processes one example at a time.

What is `batched=True`?

When `batched=True` is used, multiple examples are sent to the function at once instead of one by one.

Default batch size ≈ 1000

Example:

Before:

review1 -> function
review2 -> function

After:

[review1, review2, ... review1000] -> function

This makes the processing much faster.

Data structure when `batched=True`

With `batched=False`:

{"review": "This drug is good"}

With `batched=True`:

```
{"review": ["text1", "text2", "text3"]}
```

This means the value becomes a list of texts instead of a single text.

Tokenization Example

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("bert-base-cased")

def tokenize_function(examples):
    return tokenizer(examples["review"], truncation=True)

tokenized_dataset = drug_dataset.map(tokenize_function, batched=True)
```

Why is `batched=True` faster?

Because:

1. **Many examples are processed at the same time.**
2. **Fast tokenizers (written in Rust) use parallel processing.**

Example performance:

Option	Fast
batched=True	~10s
batched=False	~59s

What is `num_proc`?

`num_proc` **allows multiprocessing using multiple CPU processes.**

Example:

dataset.map(function, batched=True, num_proc=8)

This means 8 processes will run in parallel.

Splitting Long Text

Tokenizers can split long texts into multiple chunks.

Example:

max_length = 128

Then:

1 review → 2 or more features